

Owen Goodman

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EDUCATION

Virginia Tech <i>Master of Engineering in Computer Science and Applications</i>	Blacksburg, VA <i>Expected May 2027</i>
Virginia Tech <i>Bachelor of Engineering in Computer Science</i>	Blacksburg, VA <i>Aug 2023 - May 2026</i>

EXPERIENCE

Undergraduate Researcher <i>Virginia Tech, Dr. Luke Achenie's Lab</i>	Aug 2025 – Current <i>Blacksburg, VA</i>
- Developed LLM-based pipelines to extract explicit physical relationships from scientific literature and engineering systems - Leveraged VT ARC HPC infrastructure to train and evaluate models on large-scale STEM datasets - Contributed to research with potential publications through documentation and experimental analysis	

PROJECTS

CrateTests <i>React, Next.js, Firebase, OpenAI API</i>
- Full-stack platform converting user documents into AI-generated study materials using fine-tuned GPT-4o-mini - Built quizzes, adaptive flashcards, and personalized YouTube recommendations based on analytics - Launched subscription-based SaaS with Stripe, serving paying customers at cratetests.com
2D Isometric Game Engine <i>Java</i>
- Implemented custom 2D isometric engine with optimized tile-based rendering and camera controls - Applied architecture patterns to improve performance and maintain clean, extensible codebase - Optimized rendering pipeline to handle hundreds of isometric tiles efficiently, reducing frame lag and improving viewport management
Windows Application Usage Monitor <i>C, Win32 API, Local Web App</i>
- C application tracking active usage time for all foreground apps, storing data in JSON - Served data via local web server with interactive HTML/CSS/JS charts for visualization - Implemented high-resolution timers and efficient data structures to ensure accurate tracking with minimal performance overhead

LEADERSHIP & ACTIVITIES

Fraternity Executive Board Member <i>Kappa Alpha Epsilon Eta Chapter, Virginia Tech</i>	Nov 2024 – Nov 2025 <i>Blacksburg, VA</i>
- Managed chapter property and upheld traditions for 50+ member organization - Enforced safety protocols as sober monitor during events - Collaborated with executive team to maintain a safe and respectful community	
Varsity Soccer Captain <i>Thomas Jefferson High School</i>	Jan 2021 – Jun 2023 <i>Richmond, VA</i>
- Led varsity soccer team, fostering teamwork, discipline, and leadership on and off the field - Organized team practices and strategy sessions, improving performance and cohesion - Developed resilience and time management balancing academics and athletics	

TECHNICAL SKILLS

Languages: Python · Java · JavaScript/TypeScript · React/Next.js · C/C++ · Swift · SQL · Rust · Go
Interests: Machine Learning · AI Applications · Software Design · Game Design

RELEVANT COURSEWORK

- Data Structures & Algorithms · Intermediate Software Design · Mobile Software Development · Statistical Methods for Engineers · Applied Combinatorics · Systems · Computer Organization · CS Ethics